

A Packing of 6 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 6 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 3.054381. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.969522, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 6 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.054381 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 6$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.969522, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 6/5 = 3.054545$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N6.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
305438158632233048469807768461280338185359011453069377265145689111599655
020201963766069553032310459002214189489467961511285014609745675758241286
7168937724645060324620905605055837410433491663055
```

denominator:

```
100000000000188089367879111505366114457685216916217671556307444841333375
880401700672867668042891874471456413365916172347619106945537439329454336
1618150455535974050320184970562879919029479527707
```

B Coordinates

6 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N6.txt`.

```
723322014447 -303289799516 -291476833709 2592977434778911687045 -67742082767 27933505454 322156619561 19734474530 294125500990 -356854049295 -661952172823
109984811452 -427135698225 377794754903 -479755435638841986467 197198769738 99258552131 -383748095475-318067840388 300056733935 315849360841 226395749586
475256102756 95708502181 -81423190017 -15380547770914351249492 589285367602 -39858419682 -3402780495589555225056 21161948140 72023183004 35466071828
-77779519069 -4161272276 -366980664026 -69270352912267533719160 222202533345 -170890652326 108301392085290319954 -109904253710 -7726737902 -67860436595
-110201155500 193125747110 -596720778105 -544822821453643047985 54111345501 -262300773699 8356132486 -244277808659 405963306394 467473488462 -436225776672
391008168928 385992311012 199314802831 -295470185370104621499710 196043203274 841889460156 -1002803825558851358488 -72035244078 32618704945 -11999776314
6370187949 200703950787 395804310122 182228749057 -172935327259 423879368057 112530963754 1931814899811878875270 36397493126 360625969339 -196707537820
215151859110 58559016888 5626262962 60881549214 -578087856291 233169921779 -387600554973 -42118815424468761248 402219669334 -374951689546 -78225439475
93273449794 179995282570 229026577020 -265171862707-340741672218 193266608297 -13024305970 434516052430227889200325 -213458927324 324103946064 -53979105537
111848083121 583116369178 76409812158 37895901177 -2039274516 194730995015 -94157154798 219002658411 -236858060555 321575308715 -340035940686 382372170385
14767299099 -337596067521 142202032918 12963500531 213728524598 457402882032 -95199690447 39977903358372036417906 573457285064 238210446864 346884975399
-33839266713 251028057758 -251099035214 -46364180793018977395955 129119262685 211896600280 -1309141776281195040264 267328567464 133020025747 244445178132
110238310497 514061077275 175149215323 84273926395 134336111550 -31903021840 635534323206 -3973643832241384900017 200389816953 -194014139920 272682911507
-621672615480 177358061639 -194503374761 3345131063803296043961 187322372964 106329161119 -5763578704825857145461 -176385352750 474522757605 -205042358659
285808878405 -188737144213 423990432718 8421279009942903464127 120544202242 -359007823090 -23837518947378421899505 57138351324 497399932895 458691977133
515293914813 27486837745 135425758089 270704887684 -267376042401 -197876772856 382533208707 3947471241422810797782 -291258431745 26614381183 15248415467
24937213367 -350423573947 -128384372118 -32985449080627893666 -386266262658 95728129581 408371811127 161049351904 -441555887594 187605205723 274550809848
315949796334 240895676763 -158956992303 90756106024-463275211046 -422331635818 -438581357114 -826181404835907225886 413817106679 -166577305772 432324513906
17832967864 -440902356202 -333890344131 5894327714758700382101 -418396705085 33808346314 20664065151289781792164 212106914354 -237594209884 -142143970077
121421736811 398454859871 -224990552158 16166280440619263618 484869651025 -10546047654 616163355142 312771029895 379349388140 192717666306 -570826496976
-341994447682 -87739666797 616952604952 -3712346095477897248314 217577965071 140574903105 -209526413214369384551 120582851712 -246566314658 69645539963
143329867435 -251408402239 -292999217533 -306067919388664921527 329354567877 -205277332220 11226385287274150935651 -446676141276 -504031301838 -60366565830
```